

Acoustic imaging by three-dimensional acoustic Luneburg meta-lens with lattice columns

Cite as: Appl. Phys. Lett. **118**, 091902 (2021); <https://doi.org/10.1063/5.0037600>

Submitted: 14 November 2020 . Accepted: 17 February 2021 . Published Online: 02 March 2021

 Jung-Woo Kim, Seong-Jin Lee, Jun-Yeong Jo, Semyung Wang, and  Sang-Hoon Kim



View Online



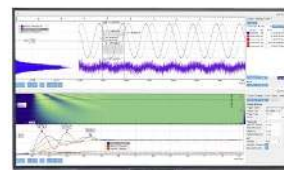
Export Citation



CrossMark

Challenge us.

What are your needs for periodic signal detection?



Zurich
Instruments



Acoustic imaging by three-dimensional acoustic Luneburg meta-lens with lattice columns

Cite as: Appl. Phys. Lett. **118**, 091902 (2021); doi: [10.1063/5.0037600](https://doi.org/10.1063/5.0037600)

Submitted: 14 November 2020 · Accepted: 17 February 2021 ·

Published Online: 2 March 2021



Jung-Woo Kim,¹  Seong-Jin Lee,² Jun-Yeong Jo,² Semyung Wang,¹ and Sang-Hoon Kim^{2,a)} 

AFFILIATIONS

¹School of Mechanical Engineering, Gwangju Institute of Science and Technology, 123 Cheomdangwagi-ro, Buk-gu, Gwangju 61005, South Korea

²Division of Marine Engineering, Mokpo National Maritime University, 91 Haeyangdaehak-ro, Mokpo 58628, South Korea

^{a)} Author to whom correspondence should be addressed: shkim@mmu.ac.kr

ABSTRACT

A three-dimensional acoustic Luneburg meta-lens has the advantage of refracting sound waves for all incident angles and focusing higher sound pressure compared to a two-dimensional lens. The lens made of plastic with a diameter of 120 mm was designed with thousands of lattice column-shaped meta-atoms to maintain its three-dimensional shape. The lens's three-dimensional focusing performance and acoustic imaging were simulated and measured in the frequency range of 5 kHz–20 kHz. The omnidirectional property was confirmed by rotating the lens to change the incident angle and measuring the sound pressure. The development of these spherical Luneburg meta-lenses is expected to improve the performance of devices that require acoustic focusing.

Published under license by AIP Publishing. <https://doi.org/10.1063/5.0037600>

The Luneburg lens, which was proposed as a model of the wave transformation theory by Luneburg in the 1940s, is a type of gradient index (GRIN) lens that can focus incoming waves on the opposite side of the lens without aberration.¹ Some theoretical and experimental validation of the lens was conducted in the 1960s,^{2,3} and the lenses with metamaterial structures were realized in various methods and shapes in the 21st century.^{4–7} Its high gain and spherical symmetry advantages, design optimization,^{8,9} printed circuit board fabrication technique,^{10,11} quasi-conformal transformation optics,¹² and metamaterial structures^{13–16} were used to apply the lenses to antennas. These studies have been mainly conducted in the electromagnetic and optical fields, but attempts have also been made to apply them to sound waves.

With the advent of metamaterials, research on the implementation of acoustic focusing^{17–23} was conducted to overcome the limitations of existing materials by controlling the waves, and the Luneburg lens was no exception. Kim realized the performance of a two-dimensional (2D) acoustic Luneburg meta-lens in the audible frequency range.^{24,25} Moreover, many methods and shapes have been proposed to achieve this. An acoustic Luneburg lens composed of orifice-type unit cells²⁶ and a 2D flattened lens designed using a quasi-conformal mapping method^{27,28} has been presented. Anisotropic acoustic Luneburg lenses such as Luneburg-fisheye combined²⁹ and Luneburg retroreflector³⁰ were introduced.

Research on the acoustic Luneburg meta-lenses has expanded to a three-dimensional (3D) shape because the 3D lenses have the advantage of focusing incident waves in all directions and achieving a higher sound pressure level (SPL) than the 2D lenses. Fabricated 2D and 3D lenses and the performance of the 2D lens in the ultrasound frequency range were studied by Xie *et al.*³¹ After that, Zhao *et al.* reported ultrasound beam steering using 2D and 3D flattened acoustic metamaterial Luneburg lenses at 40 kHz.³² Despite these pioneering studies, frequency-dependent focusing performance and acoustic imaging, consistent with 3D numerical simulation and experimental results for full 3D lenses, have not been reported.

In this paper, a 3D acoustic Luneburg meta-lens, or an acoustic Luneburg ball, composed of lattice column-shaped meta-atoms to maintain its three-dimensional shape was designed and fabricated, and its focusing performance was confirmed using COMSOL Multiphysics. Acoustic imaging was measured in the frequency range of 5 kHz–20 kHz using a transducer, microphone, three-axis linear stage, and control module. It was then demonstrated to have the omnidirectional property by rotating the lens to change the incident angle and measuring the sound pressure around the lens.

The refractive index (RI) formula of the Luneburg lens is $n(r) = n_0 \sqrt{2 - (r/R)^2}$, where n_0 is the RI of the surrounding